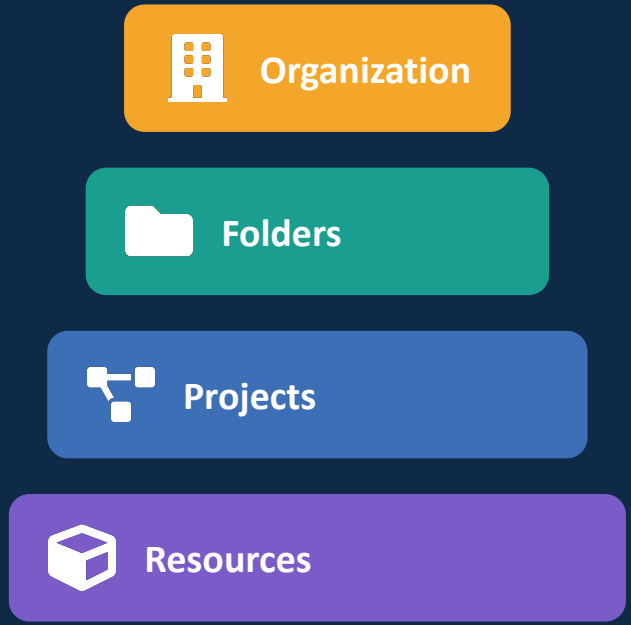
MODULE 02

Resources and Access in the Cloud

Organizing Google Cloud, controlling who can do what, and the tools you use to work with it



What we will cover today

01

Resource hierarchy

How organizations, folders, projects and resources fit together

02

Identity and Access Management

Policies that define who can do what, and where

03

IAM roles

Basic, predefined and custom roles

04

Service accounts

Identities for applications and VMs

05

Cloud Identity

Central management of users and groups

06

Interacting with Google Cloud

Console, SDK and Cloud Shell, APIs, mobile app

By the end of this module you will be able to

- Describe the four levels of the Google Cloud resource hierarchy
- Explain how policies are inherited down the hierarchy
- Choose between basic, predefined and custom IAM roles
- Explain when and why to use a service account
- Describe how Cloud Identity centralizes user management
- Pick the right tool to interact with Google Cloud for a task



Module format

6 topics

12-question knowledge check

1 hands-on lab

01

Resource hierarchy

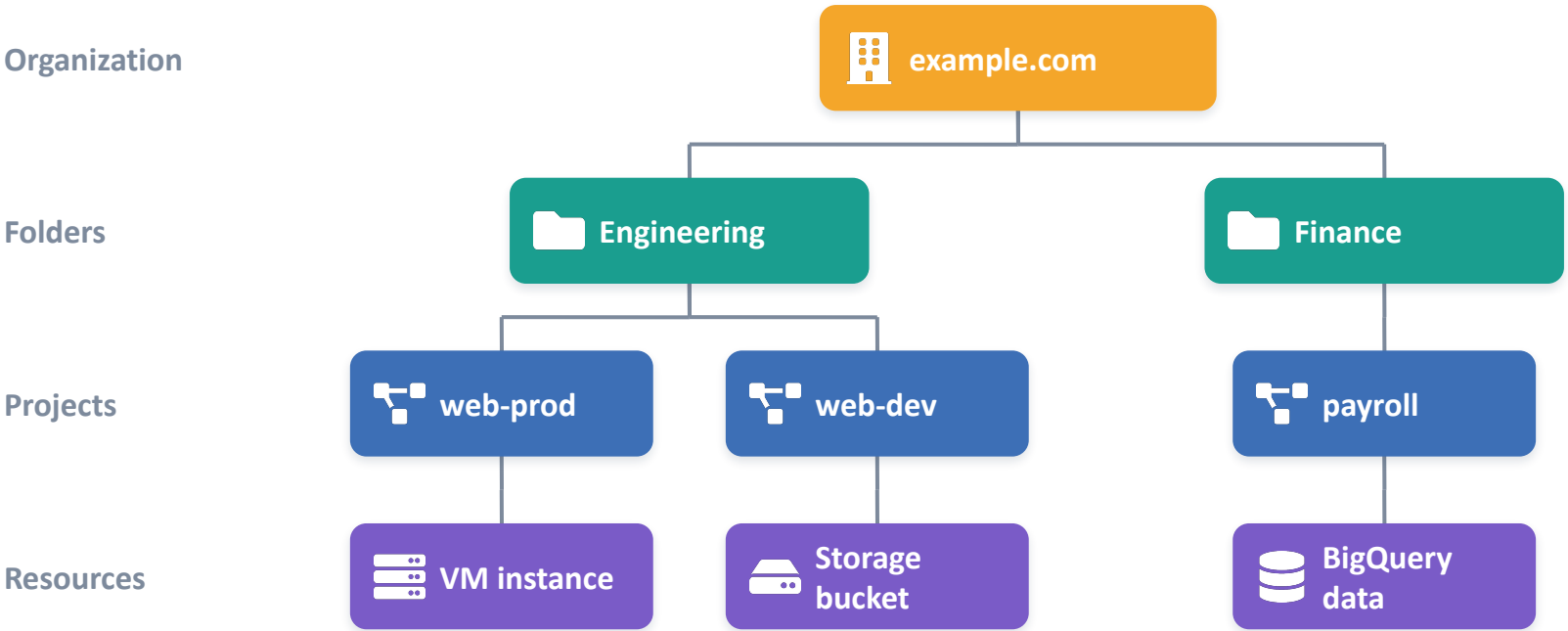


How Google Cloud organizes everything you build – and why that structure drives access and policy

Think of it like a company building

CLOUD LEVEL	REAL-WORLD ANALOGY
 **Organization node**	The whole company headquarters
 **Folders**	Departments or floors (Finance, Engineering)
 **Projects**	Individual team offices, each with its own budget
 **Resources**	The equipment inside each office (VMs, buckets, datasets)

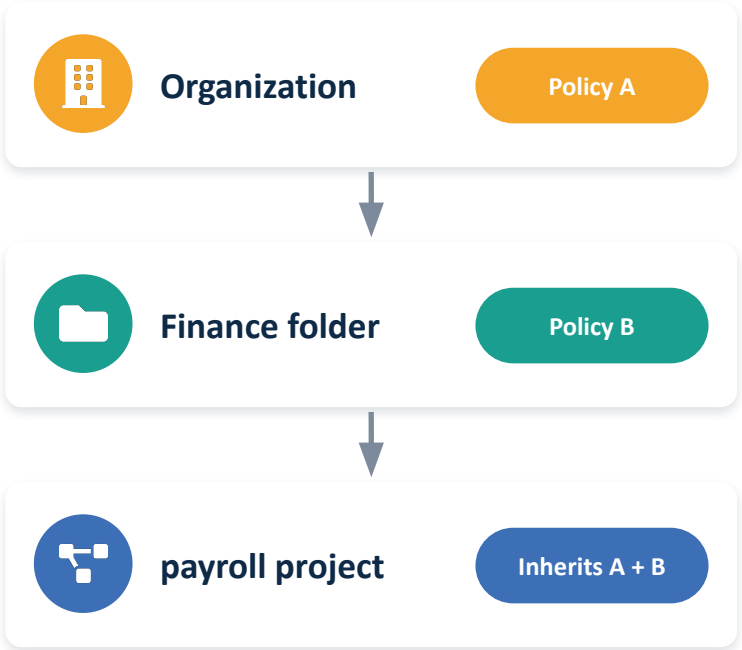
The four levels in practice



Read bottom-up: resources live in projects, projects can be grouped in folders, and everything rolls up to one organization node.

Policies flow downward

- Policies can be attached at the organization, folder and project levels – and some services also allow them on individual resources
- A policy set on a parent automatically applies to every child below it
- A resource's effective policy = its own policy plus everything inherited from above



Set a rule once on the Finance folder and every Finance project picks it up automatically.

Projects: where the work happens



Turn on services

Enable and manage the APIs and services the team needs



Contain resources

Every resource belongs to exactly one project



Own access list

Each project can have its own owners and users



Bill separately

Costs are tracked and billed per project

A project is a self-contained compartment: its own services, its own people and its own bill.

Every project has three identifiers

	Project ID	Project name	Project number
Example	`my-web-app-4821`	My Web App	`415104041262`
Who sets it	Google suggests one; you can customize it at creation	You	Google Cloud
Unique?	Yes, globally	No	Yes, globally
Can it change?	**No – fixed after creation**	**Yes, any time**	**No**
Mainly used by	You and tools, to target a project	People, for readability	Google Cloud internally

Resource Manager: manage projects with code



Resource Manager

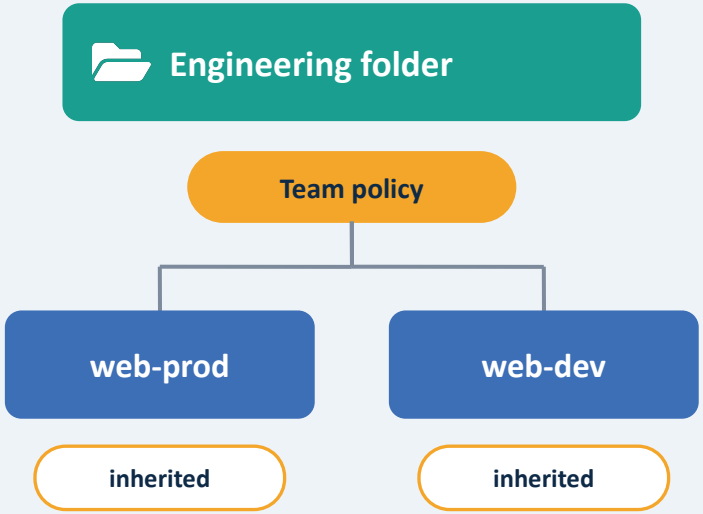
An API for working with projects programmatically, instead of clicking through the console

Reachable via RPC and REST APIs

- List all projects for an account
- Create new projects
- Update existing projects
- Delete projects
- Recover deleted projects
- Automate in pipelines and scripts

Folders: group projects and delegate control

- A folder can hold projects, other folders, or both
- Group by department, team or environment
- Delegate admin rights so teams can work independently
- Attach a policy once; every project inside inherits it



Same team, same rules, one place to manage

Without folders

Copy the same policy onto each project

With folders

Set it once on the folder; change it in one place

The organization node sits at the top

SPECIAL ROLES AT THIS LEVEL



Organization policy administrator

Only privileged people can change organization-wide policies



Project creator

Controls who can create projects – and therefore who can spend money

Folders require an organization node – it is the root that every project and folder hangs from.

HOW DO YOU GET ONE?

Already a Google Workspace customer?

Your projects automatically belong to your organization node

Not using Workspace?

Create the organization node through Cloud Identity



A new organization node still lets everyone in the domain create projects and billing accounts – tighten this deliberately.

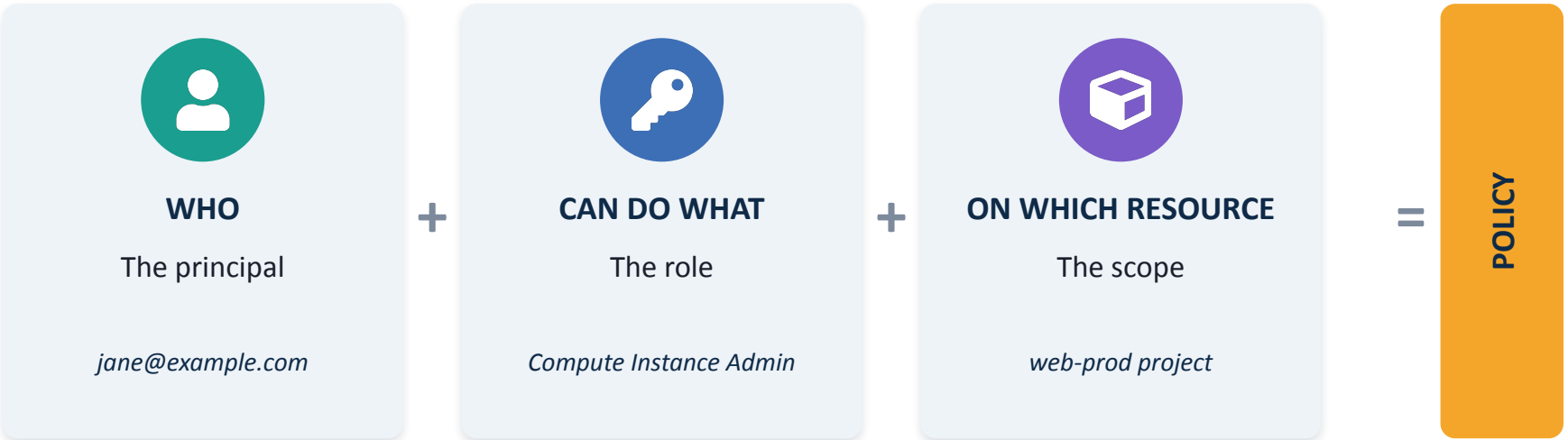
02

Identity and Access Management

Defining who can do what, on which resource – and how to block access with deny policies



An IAM policy answers three questions



Example: "jane@example.com can manage VMs in the web-prod project."

Principals and roles

WHO CAN BE A PRINCIPAL



Google account

A single person



Google group

A set of people



Service account

An app or VM



Cloud Identity domain

Everyone in a domain

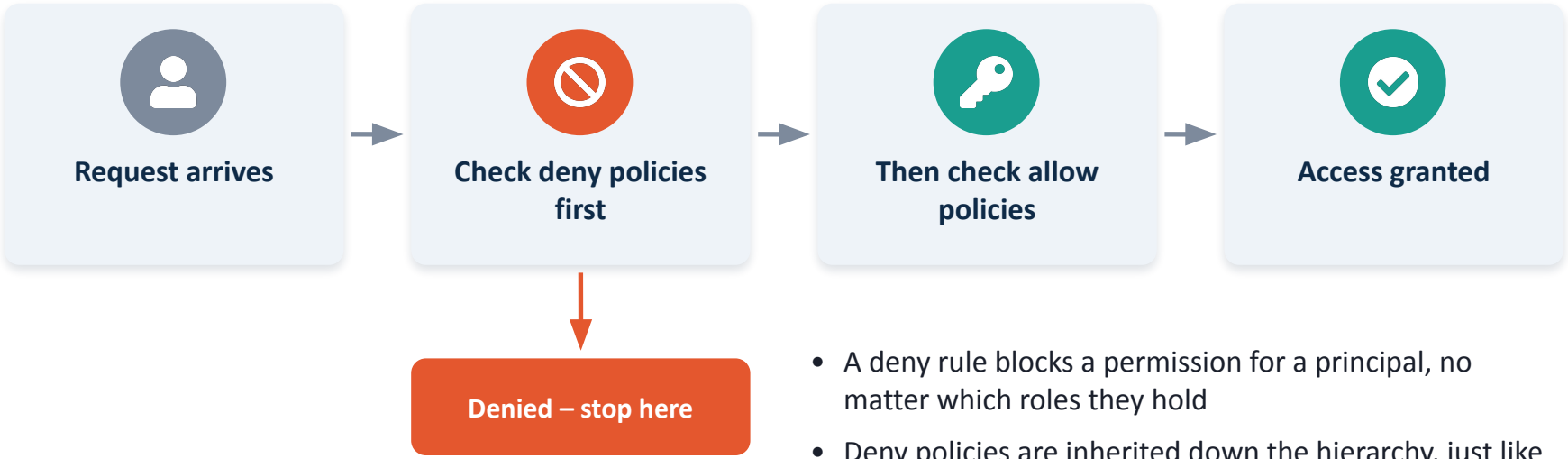
A ROLE IS A BUNDLE OF PERMISSIONS

To manage VMs, a person needs to…

- ✓ create virtual machines
- ✓ delete virtual machines
- ✓ start virtual machines
- ✓ stop virtual machines
- ✓ change virtual machines

…so they are grouped into one role you can grant in a single step.

Deny policies override allow policies



- A deny rule blocks a permission for a principal, no matter which roles they hold
- Deny policies are inherited down the hierarchy, just like allow policies

03

IAM roles



Three role types, from broad to precise – and how to pick the right one for least privilege

Three types of role, from broad to precise

Broadest

Finest-grained





Basic

Owner, Editor, Viewer, Billing Admin

Apply to every resource in the project

Maintained by: Google



Predefined

Service-specific, e.g. Compute Instance Admin

Match typical job needs for one service

Maintained by: Google



Custom

Exactly the permissions you list

Fit least-privilege needs precisely

Maintained by: You

Basic roles: quick to grant, very broad

Role	View resources	Change resources	Manage access	Manage billing
Viewer	✓	–	–	–
Editor	✓	✓	–	–
Owner	✓	✓	✓	✓
Billing Administrator	–	–	–	✓

Billing Administrator lets someone manage costs without touching any resources.



Use with care

On a project with sensitive data, basic roles usually grant far more than anyone needs. Prefer predefined or custom roles.

Predefined vs custom roles in action



instanceAdmin

Predefined | defined by Google

- ✓ compute.instances.get
- ✓ compute.instances.list
- ✓ compute.instances.start
- ✓ compute.instances.stop
- ✓ compute.instances.delete
- ✓ compute.instances.setMachineType



instanceOperator

Custom | defined by you

- ✓ compute.instances.get
- ✓ compute.instances.list
- ✓ compute.instances.start
- ✓ compute.instances.stop
- ✗ ~~compute.instances.delete~~
- ✗ ~~compute.instances.setMachineType~~

Where custom roles can be created

Organization	Allowed	Can include any permission supported in custom roles
Folder	Not allowed	Need one for a folder? Define it at the organization level instead
Project	Allowed	Any supported permission except org- or folder-only ones, such as resourcemanager.organizations.get

Remember

Custom roles give precise least privilege, but you own their upkeep as permissions evolve.

Tip: start from a predefined role and remove what you don't need.

04

Service accounts



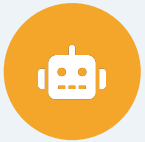
Giving applications and virtual machines their own identity – without sharing anyone's password

Let the workload act on its own behalf



App on a VM

Needs to save files regularly



Service account

Identity granted a storage role



Cloud Storage

Accepts writes only from that identity

Without a service account

A person grants access each run, or credentials get pasted into code – slow and risky.

With a service account

The VM itself holds just the permissions it needs, with no human in the loop.

Service accounts: identity and resource



As an identity

`uploader@my-proj.iam.gserviceaccount.com`

- Named with an email address
- Authenticates with cryptographic keys, not passwords
- Granted roles like any principal – e.g. with Instance Admin, its app can create, modify and delete VMs



As a resource

It can have IAM policies of its own, controlling who manages it:



Alice – Editor on the service account

Manages who can act as it



Bob – Viewer on the service account

Can only list and view it

05

Cloud Identity



Moving from scattered personal accounts to centrally managed users, groups and policies

From quick start to central control

HOW MANY TEAMS START



Personal Gmail accounts plus Google Groups

- Easy to set up on day one
- Identities aren't centrally managed
- When someone leaves, there's no quick way to cut their access

WITH CLOUD IDENTITY



Users, groups and policies in the Google Admin console

- One place to manage every identity
- Define organization-wide policies
- Disable a leaver's account in minutes

What Cloud Identity gives you



Reuse existing logins

Sign in with the same credentials already used in Active Directory or LDAP



Offboard instantly

Disable accounts and remove leavers from groups in the Google Admin console



Free or Premium

Premium adds mobile device management



Built in for Workspace

Workspace customers already have this in the Admin console

06

Interacting with Google Cloud

Four ways to work with your resources – from point-and-click to fully automated



Four ways to work with Google Cloud

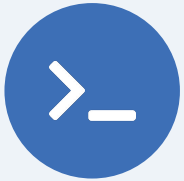
01



Google Cloud console

Web-based graphical interface

02



Cloud SDK and Cloud Shell

Command-line tools

03



APIs

Control services from your own code

04

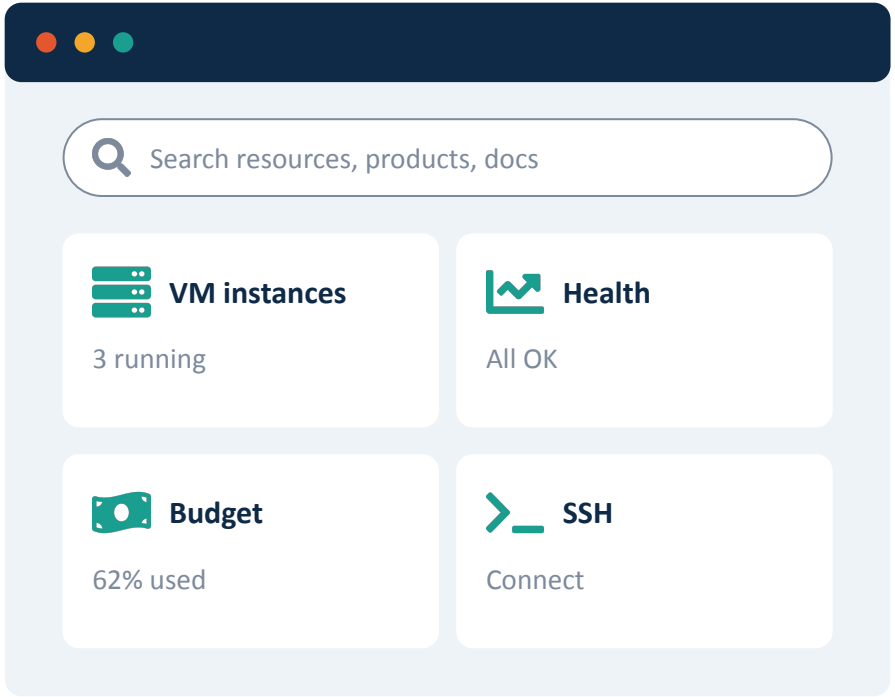


Google Cloud app

Manage resources from your phone

Same resources, same IAM permissions – just different interfaces for different tasks.

Google Cloud console: the web interface



- Deploy, scale and diagnose production issues from a browser
- Find resources, check their health and manage them fully
- Set budgets to keep spending under control
- Search for anything and SSH into instances right in the browser

Cloud SDK and Cloud Shell: the command line

```
$ gcloud config list
project = my-web-app-4821

$ gcloud compute instances list
NAME     ZONE           STATUS
web-1    us-central1-a  RUNNING

$ bq ls
  sales_data
```

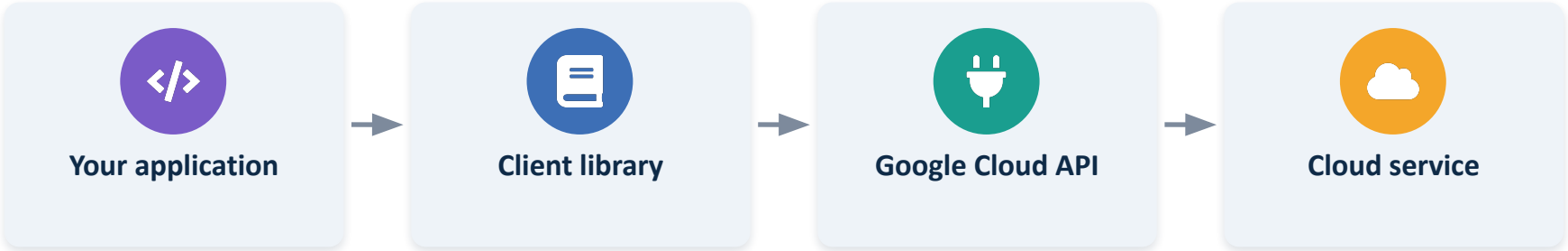
Google Cloud SDK

- gcloud CLI: the main command line for Google Cloud
- bq: a dedicated tool for BigQuery
- Installed tools live in the SDK's bin directory

Cloud Shell

- Command line in your browser – nothing to install
- Debian-based, with a persistent 5 GB home directory
- Pre-authenticated, SDK pre-loaded, always up to date

APIs: control Google Cloud from your own code



Google APIs Explorer

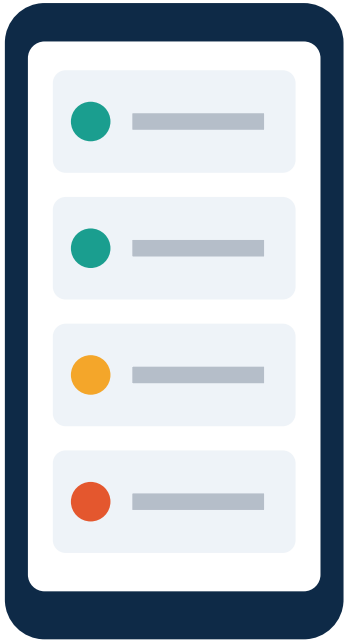
Built into the console: see which APIs exist, their versions, and try them interactively – even ones needing sign-in.

Client libraries take the drudgery out of calling APIs, in:

- Java
- Python
- PHP
- C#
- Go
- Node.js
- Ruby
- C++

Google Cloud app: manage on the move

- Start, stop and SSH into Compute Engine VMs, and view their logs
- Stop and start Cloud SQL instances
- Administer App Engine apps: view errors, roll back, split traffic
- Up-to-date billing, with alerts for projects going over budget
- Custom graphs for CPU, network, requests per second, errors
- Alerts and incident management



cloud.google.com/app

Which tool would you reach for?

SCENARIO	BEST FIT
You're on a train and get a budget alert	**Google Cloud app**
Run one gcloud command without installing anything	**Cloud Shell**
Explore a new service visually and set a budget	**Cloud console**
Your Python app must create projects every night	**APIs + client library**
Script BigQuery loads from your laptop	**Cloud SDK (bq)**

Key takeaways

1

Hierarchy

Organization > folders > projects > resources; every resource lives in one project

2

Inheritance

Policies flow down the tree; set them as high as makes sense

3

IAM

Who + role + resource; deny rules are checked before allow rules

4

Roles

Basic is broad, predefined fits jobs, custom fits least privilege

5

Service accounts

Identities for workloads that use keys, not passwords

6

Tools

Console, SDK and Cloud Shell, APIs, and the mobile app

?

Knowledge check



12 questions covering all six topics. Answer each one before we reveal it.

Question 1 of 12

Resource hierarchy

Which list shows the resource hierarchy from top to bottom?

A Projects, folders, organization, resources

B Organization, folders, projects, resources

C Organization, projects, folders, resources

D Folders, organization, projects, resources



Hint (if needed): Think of the company building: HQ first, equipment last.

Answer 1: B

Resource hierarchy

Which list shows the resource hierarchy from top to bottom?

A Projects, folders, organization, resources

✓ **Organization, folders, projects, resources**

C Organization, projects, folders, resources

D Folders, organization, projects, resources

Why

The organization node is the root. Folders (optional) group projects, and projects contain the resources.

Question 2 of 12

Projects

Your team needs to turn on the BigQuery API. At which level are APIs and services enabled?

A Organization

B Folder

C Project

D Individual resource



Hint (if needed): Which level has its own services, people and bill?

Answer 2: C

Projects

Your team needs to turn on the BigQuery API. At which level are APIs and services enabled?

A Organization

B Folder

✓ **Project**

D Individual resource

Why

APIs and services are enabled on a per-project basis. Each project chooses which services it uses.

Question 3 of 12

Project identifiers

Which project identifier can you change at any time after the project is created?

A Project ID

B Project number

C Project name

D None of them can change



Hint (if needed): One of them is just a friendly label.

Answer 3: C

Project identifiers

Which project identifier can you change at any time after the project is created?

A Project ID

B Project number

✓ **Project name**

D None of them can change

Why

The project name is a human-readable label you choose; it need not be unique and can be changed whenever you like.

Question 4 of 12

IAM role types

Which order runs from the broadest IAM role type to the most fine-grained?

A Custom, predefined, basic

B Predefined, basic, custom

C Basic, predefined, custom

D Basic, custom, predefined



Hint (if needed): Which type affects every resource in a project?

Answer 4: C

IAM role types

Which order runs from the broadest IAM role type to the most fine-grained?

A Custom, predefined, basic

B Predefined, basic, custom

✓ **Basic, predefined, custom**

D Basic, custom, predefined

Why

Basic roles cover every resource in a project, predefined roles target one service's job needs, and custom roles contain exactly the permissions you list.

Question 5 of 12

Basic roles

A finance colleague must manage the project's billing but must never change any resources. Which basic role fits?

A Owner

B Editor

C Viewer

D Billing Administrator

Hint (if needed): One basic role is all about money and nothing else.

Answer 5: D

Basic roles

A finance colleague must manage the project's billing but must never change any resources. Which basic role fits?

A Owner

B Editor

C Viewer

✓ **Billing Administrator**

Why

Billing Administrator controls billing for a project without granting permission to change its resources.

Question 6 of 12

Deny policies

A user holds Editor on a project, but a deny policy blocks them from deleting VMs. What happens when they try?

A The delete succeeds, because allow policies win

B The delete is blocked, because deny is checked first

C It depends on which policy was created most recently

D It works only through the console



Hint (if needed): Which policy type does IAM evaluate first?

Answer 6: B

Deny policies

A user holds Editor on a project, but a deny policy blocks them from deleting VMs. What happens when they try?

A. The delete succeeds, because allow policies win

✓ **The delete is blocked, because deny is checked first**

C. It depends on which policy was created most recently

D. It works only through the console

Why

IAM evaluates deny policies before allow policies. A matching deny rule blocks the action regardless of the roles granted.

Up next: hands-on lab



Getting started with Cloud Marketplace

Use Cloud Marketplace to launch a complete Linux web stack (LAMP) on a Compute Engine VM in just a few clicks.

1. Sign in to the lab project in the console
2. Find the LAMP stack in Cloud Marketplace
3. Deploy it to a Compute Engine instance
4. Open the site and confirm it works

THANK YOU

Questions and discussion

Resource hierarchy | IAM | Roles | Service accounts | Cloud Identity | Tools

